Advanced Workflows

Combining dynamic reporting with version control, automation, and containerization

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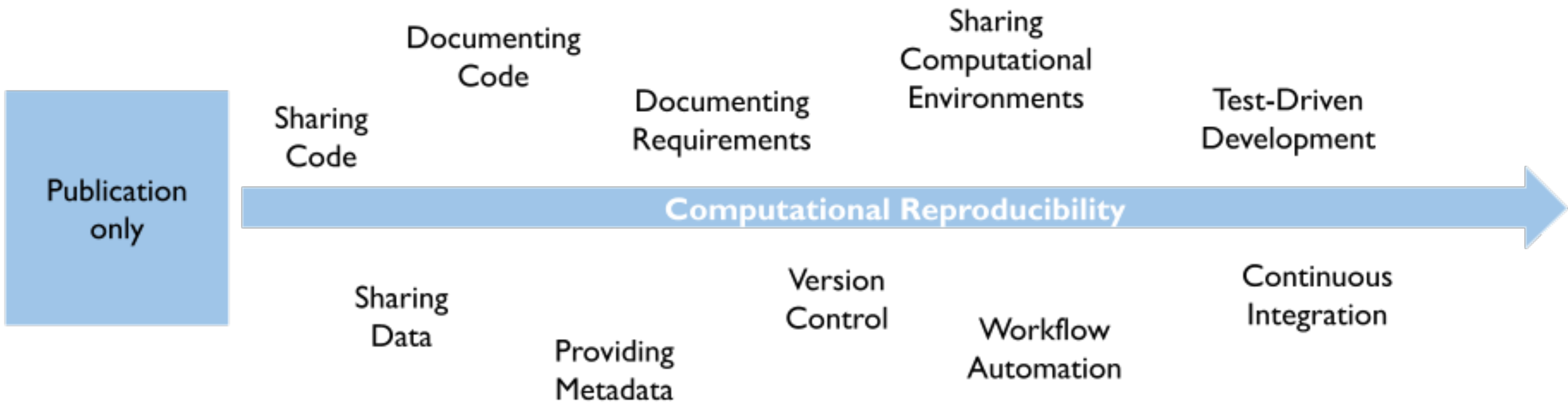
Reproducibility is a spectrum



- Goal today: Understanding **what is possible** towards the right side
- ~~Goal today: Deep technical understanding of advanced workflows~~

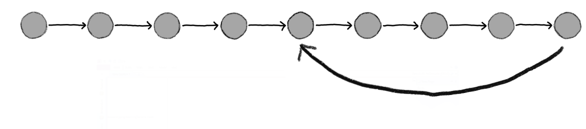
Dynamic reporting solves copy-paste problem, but …

- How to keep track of different **file versions**?
  - `paper-final.qmd` → `paper-final2.qmd` → `paper-final2b.qmd` → 😱
  - Solution: **Git**
- How to know which files to run in which **order**?
  - `functions.qmd`, `analysis.qmd`, `plots.qmd`, `paper.qmd`, 🤔
  - Solution: **Make**
- How to make sure code still runs when **software versions** change?
  - `Today`: R version 4.5, ggplot2 3.5.2 → `Next year`: R version ❓, ggplot2 ❓
  - Solution: **Docker**
- Very nice paper on such a workflow: Peikert and Brandmaier (2021)

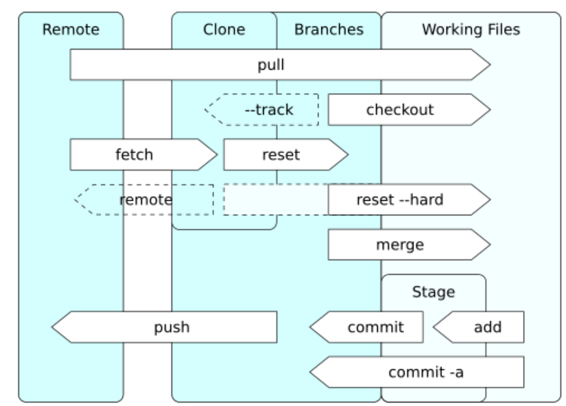
Version control with Git

- **De facto standard** version control system
- Can **go back in time** to previous versions
- Can **track changes** between versions
- Can **branch and merge**
- Useful for **collaboration** and **file transfer**
- **Command line** interface (learning curve)
- **Graphical interface** in many editors/IDEs (e.g., RStudio, VS Code)
- **Git** (software) ≠ **GitHub/GitLab** (Git repository hosting services)
- Snapshots enable **persistently archiving** (with DOI) on data repositories (e.g., Zenodo)

https://git-scm.com/downloads/logos



The Turing Way Community and Scriberia (2024)



https://commons.wikimedia.org/wiki/File:Git_operations.svg

Automation with Make

- Tool for managing **computational recipes**
- **Automating multiple steps** into one

> Arrabbiata sauce, or sugo all'arrabbiata in Italian, is a spicy sauce for pasta made from garlic, tomatoes, and dried red chili peppers cooked in olive oil.

https://en.wikipedia.org/wiki/Arrabbiata_sauce

Makefile

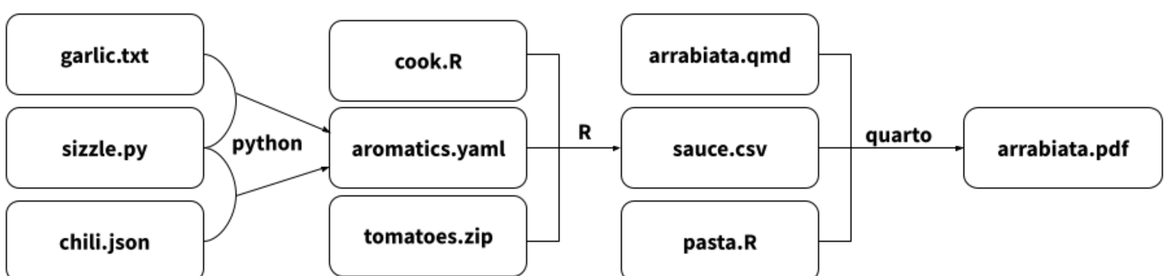
```
arrabiata.pdf: arrabiata.qmd sauce.csv pasta.R
  quarto render arrabiata.qmd

sauce.csv: cook.R tomatoes.zip aromatics.yaml
  R -e "source(cook.R)"

aromatics.yaml: sizzle.py garlic.txt chili.jso
  python sizzle.py garlic.txt
  python sizzle.py chili.json
```

Terminal

```
make arrabiata.pdf # execute recipe
```



Containerization with Docker

- **Software containers** encapsulate computational environment
- Computational environment components:
  - Operating system
  - R and R packages
  - Numerical software libraries (e.g., LAPACK, BLAS)
  - other software (e.g., LaTeX, Java)
- **Docker** is a popular tool for containerization
- **Rocker** is a Docker image for R (https://rocker-project.org/)

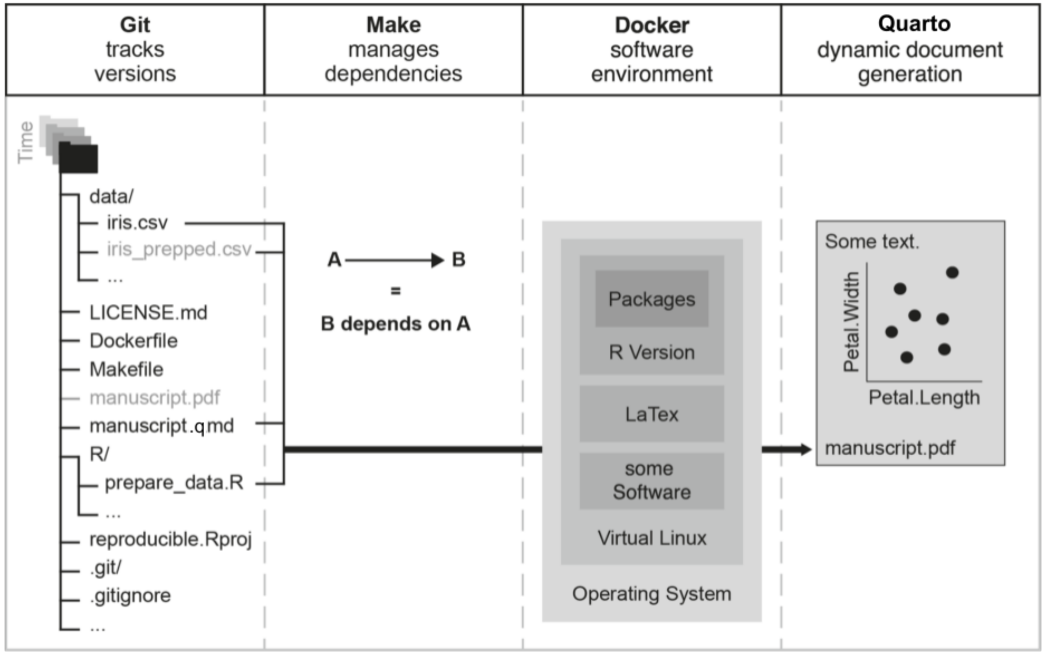
https://commons.wikimedia.org/wiki/File:Docker-svgrepo-com.svg

Dockerfile

```
# Uses a Rocker container image with R version 4.4
FROM rocker/verse:4.4

# Install R package versions at time when R 4.4 was up-to-date
RUN install2.r ggplot2 dplyr kableExtra
```

Putting it all together



Terminal

```
# clone git repository
git clone git@gitlab.uzh.ch:crsuzh/zurich-names.g

# enter project repository
cd zurich-names

# run automated analyses
make docker-report-html
```

→ `report.html`

https://gitlab.uzh.ch/crsuzh/zurich-names

Modified from Peikert and Brandmaier (2021)

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Conclusions

- Dynamic reporting + version control + automation + containerization
  - improve **reproducibility**, **reuseability**, **collaboration**
  - lends research more **transparency**, **credibility**
- There is a **learning curve** 😨, but …
  - start simple and learn **step-by-step** 💡
  - tools are **constantly evolving** 🚀
  - invested learning time can **save you time** in the future ⏳
  - skills are **useful in/outside Academia** 💰 👨‍🔬

The Swiss Reproducibility Network Academy

- **Early career** researchers section of the SwissRN
- Goal 1: Connect young researchers interested in **reproducibility in Switzerland**
- Goal 2: Improve reproducibility of research in Switzerland
- How to join?
  - young researchers in **Switzerland** (working language: English)
  - **interest** in improving research and reproducibility
  - **every field** is welcome (diversity is the key!)
- More information: https://www.swissrn.org/contents/academy/

References

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